

CloudFormation Template Upload

Automatic Upload to S3

To automatically upload the enhanced CloudFormation template to your S3 bucket:

```
# Run the upload script  
./scripts/upload-cfn-template.sh
```

This will: 1. Validate the CloudFormation template syntax 2. Upload to s3://cloudguard-cfn-templates/cloudformation/aws-readonly-role.yaml 3. Set public read permissions 4. Display the public S3 URL 5. Keep created AWS resources tagged with Owner=cloudvul@iitm

Manual Upload (Alternative)

If you prefer to upload manually:

```
# Upload to S3  
aws s3 cp infra/enhanced-cloudformation-template.yaml \  
    s3://cloudguard-cfn-templates/cloudformation/aws-readonly-role.yaml \  
    --region eu-north-1 \  
    --content-type "text/yaml" \  
    --acl public-read  
  
# Verify upload  
aws s3 ls s3://cloudguard-cfn-templates/cloudformation/
```

Template URL

After upload, the template will be available at:

<https://cloudguard-cfn-templates.s3.eu-north-1.amazonaws.com/cloudformation/aws-readonly-role.yaml>

CI/CD Integration

Add to your deployment pipeline:

```
# GitHub Actions example  
- name: Upload CloudFormation Template  
  run: ./scripts/upload-cfn-template.sh  
  env:  
    AWS_ACCESS_KEY_ID: ${ secrets.AWS_ACCESS_KEY_ID }  
    AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }  
    AWS_DEFAULT_REGION: eu-north-1
```

Troubleshooting

Permission Denied

```
chmod +x scripts/upload-cfn-template.sh
```

AWS Credentials Not Configured

```
aws configure
```

```
# Enter your AWS Access Key ID, Secret Access Key, and region
```

S3 Bucket Doesn't Exist

```
# Create the bucket
```

```
aws s3 mb s3://cloudguard-cfn-templates --region eu-north-1
```

```
# Tag the bucket
```

```
aws s3api put-bucket-tagging \  
  --bucket cloudguard-cfn-templates \  
  --tagging '{"TagSet":[{"Key":"Owner","Value":"cloudvul@iitm"}]}'
```

```
# Enable public access (if needed)
```

```
aws s3api put-public-access-block \  
  --bucket cloudguard-cfn-templates \  
  --public-access-block-configuration \  
    "BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublic
```